THE EMPLOYEE NEED

From procedure search to guidance you can verify.

A working conversational assistant for the situation behind a branch employee's question.

EMPLOYEE TASK

Find the applicable rule, conditions and exceptions.

DESIGN RESPONSE

Ask → inspect the source → refine the situation.

A REAL PROCEDURE QUESTION

How much coin can stay in a machine overnight?

BRANCH ASSIST / RECORDED ANSWER

$5,250 maximum overnight in a single machine

Open the cited rule and inspect its conditions.

PR63 · original source available

11 supplied procedures **139** HTML tables **1,388** traceable passages

Business goal to test: faster access to correct guidance, with evidence employees can inspect.

Interview prototype · supplied Cayman Bank snapshot · AI-assisted development and runtime answers · no transaction execution

Reliable answers start with well-organized rules.

Preserve the situation attached to each instruction before searching for an answer.

PR63: keep the task and control together

Situation	Control
Monthly testing	**Single**
Weekly balancing	**Dual**
Random monthly cash count	**Dual**

01 / PARSE

Keep structure

HTML headings, lists and table rows. Linked content is not fetched.

02 / CHUNK

Keep context

~110 narrative words; 20-word overlap. Carry table labels and conditions.

03 / TRACE

Keep identity

Procedure, section, date and original-file fingerprint on each passage.

SEARCH THE SAME VERIFIED PASSAGE COLLECTION

Exact terms
BM25 keyword search

›

Same meaning
MiniLM · 384 dimensions

›

Combine + rerank
Rank fusion + cross-encoder

›

Select evidence
≤3 passages / ~520 words

Supplied PR63 facts · 110 words is a narrative target, not an exact token count · table continuations repeat their condition

Retrieve the rule. Generate an answer. Check the claims before release.

A small local pipeline makes evidence, control decisions and failure outcomes inspectable.

PREPARED OFFLINE Verified passages + keyword statistics + normalized vectors

UNDERSTAND	RETRIEVE	SELECT	GENERATE	CHECK	RESPOND
Question context + search scope	BM25 + MiniLM fusion + rerank	≤3 passages ~520 words	Qwen3-4B Q4 local CPU	Citations Numbers Support	Cited answer or source review

Weak evidence → clarify or report insufficient support.

Failed checks → one revision → withhold if still failing.

Why retrieval-augmented generation?

Replaceable source material and explicit citations.

Why hybrid search?

Exact identifiers plus paraphrases; jointly rerank.

Why this runtime?

Local CPU, no API key; small in-memory index.

MiniLM-L6 embeddings/reranker · DeBERTa NLI support screen · bounded orchestration · no transaction tools or confidence percentage

One question becomes a conversation you can inspect.

Use the source to check the answer, then change the task or refine the situation.

RECORDED TOOL WORKFLOW / PR63

What is the overnight storage limit for a coin-counting machine?

The overnight storage limit for a coin-counting machine is $5,250 [S1].

And is monthly testing done in single or dual control?

Monthly testing is done in single control. [S2]

[S2] PR63 / MONTHLY TESTING

Coin-counting machines

Monthly testing

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Source file matches the supplied snapshot

> The employee assigned to complete the single control Coin-Counting Machine Testing Control Online task must complete the monthly testing before the coin machine is used in the morning or just before balancing.
> Obtain the following coin from either an employee cashline or the vault cashline:

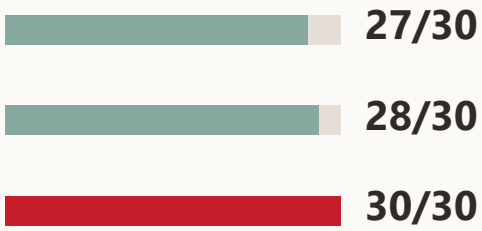
Same procedure. Fresh evidence for the new task.

Actual recorded answers and source-pane capture · inspect Evidence / Checks / Activity · topic switch, reset, feedback and export are supported

Finding evidence is the first test. Answer quality is the next.

Measured local regression results; independent review is still required for an operational decision.

RELEVANT PASSAGE IN TOP FIVE



23/30 answers pass the expected-fact proxy

Phrase matching can miss valid wording or pass an incorrect answer.

WHAT TO MEASURE / WHAT A REVIEWER MUST JUDGE

Accuracy	Citations, numbers and support screens.	Is every instruction correct for this situation?
Relevance	Passage ranking and direct response.	Does it answer the exact question?
Completeness	23/30 expected-fact proxy.	Are material conditions and exceptions included?
Conciseness	28-word mean draft; bounded output.	Is it brief without losing necessary guidance?

Next: two independent reviewers, unseen scenarios, critical-error checks.

6/35 generated evaluations: source only

2/4 original follow-ups: fact-proxy pass

CPU p50 / p95: 13.8s / 23.9s

30 procedure questions + 4 two-turn conversations + 6 selected guardrails · AI-authored/source-checked labels · cold load excluded · draft length n=32

Demonstrate the workflow. Validate before a pilot.

The working prototype is ready to inspect; the next decision is a scoped validation phase.

DEMONSTRATE / LIVE TOOL OR APPROVED RECORDING

01 Ask a routine rule

Coin overnight limit → monthly-testing follow-up.

$5,250 overnight; single control for testing. [PR63]

02 Change a condition

Emailed notary document → requester is present.

The emailed-document restriction still applies. [PR1000]

03 Inspect an exception

Newer account → Private Bank Experience exception.

$1,000/day rule; review the source if checks fail. [PR1949]

VALIDATE / OWNERS AND GATES

Quality · procedure owners + SMEs

Unseen scenarios; four quality scores.
Critical errors and coverage are separate gates.

Value · branch operations + AI team

Task success; time to correct guidance.
Compare with current search.

Pilot controls · security + platform

Identity, source access/versioning, monitoring and rollback.

Recommendation: validate quality and employee value before an operational pilot.

20-minute presentation → 15-minute walkthrough · approved silent recording 9:25 + live inspection/Q&A · AI assistance disclosed; candidate owns decisions